HTML5

24 February 2026 20:22

What is HTML?
HTML(HyperText Markup Language) is a standard language to create web pages.
It tells the browser:

- What content to display
- How content is structured
- What each part of the page means

HTML is not a programming language -- it is a markup language.

What is HTML5?
Is the latest version of HTML used to design modern website and web applications.
It improves:

- Multimedia support
- Structure of web pages
- Performance
- Mobile compatibility
- Interactive features

HTML5 removes the need for many external plugins.

Real life Analogy
Think of a web page as a classroom notice board:

Part	Real-life equivalent
HTML	Structure of the board
CSS	Decoration
JavaScript	Interactivity

Declares HTML5 document

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h1>Basic structure of an HTML5 page</h1>
    <h2>Declares HTML5 document</h2>
    <h3>Root element</h3>
    <h4>Hello</h4>
    <h5>World</h5>
    <h6>Hi!</h6>
</body>
</html>
```

Root element

english.

Page information.

Browser tab title.

Visible content

Heading are of different types.

```
    <h6>Hi!</h6>
</body>
</html>
```

Heading are of different types.

'/' is used to end the tag.

Important feature of HTML5

- **Semantic Tags(Meaningful Structure)**

HTML5 introduces tags that describe content clearly.
Example:

```
<header>Website header</header>
<nav>Navigation menu</nav>
<section> Article content</section>
<footer>Footer area</footer>
```

Helps search engine understand content.

- **Multimedia support**

```
<audio src="music.mp3"></audio>

<video src="video.mp4"></video>
```

- **Forms with new input types**

```
<input type="email">
<input type="date">
<input type="number">
<input type="range">
```

Benefits:

- **Automatic validation**
- **Better user experience**

- **Canvas for Graphics**

HTML5 allows drawing graphics directly.

```
<canvas id="mycanvas" width="200" height="100"></canvas>
```

Used for:

- Games
- Charts
- Animations

Advantages of HTML5

- **Mobile friendly**
- **Cleaner code structure**

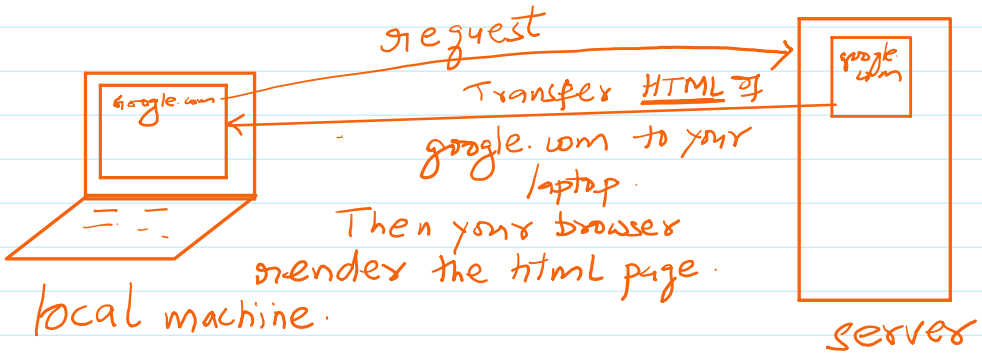
- **Faster websites**
- **Built-in multimedia**
- **Better accessibility**
- **SEO(Search Engine Optimizer) Friendly**

Example code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>SigmaPi Academy</title>
</head>
<body>
    <header>
        <h1>Welcome Students</h1>
    </header>
    <section>
        <p>
            Learning HTML5 is the first step to web development.
        </p>
    </section>
    <footer>
        <p>&copy SigmaPi Academy</p>
    </footer>
</body>
</html>
```

Homework:

1. What is HTML5?
2. Why is <doctype> is important?
3. What are semantic tags?
4. Advantages of HTML5 multimedia?



(*) GIT (versioning system) ← software

* GIT (versioning system)
* Create a github account. (github.com)
  service provider.